

Memory Interfacing: RAM & ROM Decoding — MPCA

Module 4

Definition: Memory Interfacing is the electronic hardware arrangement that connects semiconductor memory chips (RAM and EPROM/ROM) to the microprocessor's Address, Data, and Control buses using **Address Decoding Logic** to assign unique physical address ranges to each memory chip.

1. Detailed Technical Explanation

1. 8086 Memory Organization Review

- The 8086 microprocessor has a **20-bit Address Bus (A₀ - A₁₉)**, enabling it to address up to **1 MB (2²⁰ bytes)** of physical memory.
- Physical memory is organized into two 512 KB banks:
- Even Bank (Lower Bank):** Connected to Data Bus lines **D₀ - D₇**, selected by address line **A₀ = 0**.
- Odd Bank (Higher Bank):** Connected to Data Bus lines **D₈ - D₁₅**, selected by **BHE = 0** (Bus High Enable).

Microprocessor 8086		Memory Banks (1 MB Total)	
+-----+		+-----+	
	A0 = 0		EVEN BANK (512 KB RAM/ROM)
Data Lines D0-D7	=====>		Data Lines D0-D7
			+-----+
	/BHE = 0		ODD BANK (512 KB RAM/ROM)
Data Lines D8-D15	=====>		Data Lines D8-D15
			+-----+
+-----+		+-----+	

2. Address Decoding Logic (74LS138 3-to-8 Decoder)

The **74LS138** decoder is the industry standard IC used to generate chip select (**CS**) signals for multiple memory chips.

74LS138 3-to-8 DECODER			
+-----+			
A19 ----->	G1 (Active High Enable)		
A18 ----->	/G2A (Active Low)		
M//IO ----->	/G2B (Active Low)		
A17 ----->	C (Select Input 2)	----->	/Y0 (CS for Chip 0)
A16 ----->	B (Select Input 1)	----->	/Y1 (CS for Chip 1)
A15 ----->	A (Select Input 0)	----->	/Y2 (CS for Chip 2)
+-----+-----> ...			

Memory Map Design Example:

To interface a 64 KB EPROM chip (Address lines **A₀ - A₁₅** used internally inside chip):

- Higher address lines **A₁₆ - A₁₉** are connected to the decoder.
- When **A₁₉A₁₈A₁₇A₁₆ = 1111₂**, decoder output **Y₇** goes LOW, activating the EPROM chip in address range **F0000H - FFFFFH** (Reset address area of 8086).

3. Core Concepts & Memory Keywords

- **Address Decoding:** Translating upper address bus bits into unique **CS** chip select signals.
- **BHE & A₀:** Control signals used to select byte or 16-bit word memory transfers.
- **74LS138 Decoder:** 3 select inputs (**A, B, C**) to 8 active-low outputs (**Y₀ - Y₇**).

4. Must-Write Points for Exams

- ROM/EPROM is always placed at the top of memory (**FFFF0H**) because 8086 starts execution at **FFFF0H** upon reset.
- RAM is typically mapped starting from address **00000H** (to hold Interrupt Vector Table IVT).
- Both Even and Odd memory banks must receive appropriate **RD** / **WR** control strobes.

5. Quick Recall Flow

20-bit Address Bus -> Lower bits (A₀-A_k) to Memory Chip -> Upper bits to 74LS138 Decoder -> Chip Select /CS Generated